

Atakan Öztarak

Software Engineer II | MSc CS student at KIT

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Profile

Master's student at KIT with 3 years of software engineering experience specializing in backend architectures, cloud-native infrastructure, and enterprise AI engineering. Proven track record in building multi-agent systems, designing automated LLM indexing pipelines (RAG), and optimizing repository-scale code intelligence platforms. Experienced across fintech, genomics, and firmware delivery workflows.

Education

Karlsruhe Institute of Technology (KIT)

Present | Karlsruhe, Germany

Master of Science - Computer Science

- Area of specialization: Human-centered Machine Intelligence, Telematics

Middle East Technical University (METU)

Ankara, Turkey

Bachelor's - Computer Engineering

- Student Assistant for CENG240: Programming with Python for Engineers
- Graduated with **Honors**
- University Placement Exam: Ranked in the top **0.04%** nationally

Professional Experience

Software Engineer (Werkstudent) - IBM Z Firmware

Mar 2026 – Present

Delivery

Stuttgart, Germany

IBM

- Engineered an automated AI data pipeline to parse and index large-scale software repositories with LLMs via RAG.
- Designed and implemented a multi-repository connection framework that allows AI code editors to trace dependencies and maintain deep contextual awareness across distributed codebases.
- Architected semantic search and embedding strategies to optimize context window utilization, significantly improving the precision of code generation and codebase explanation tools.
- Deployed and orchestrated the AI pipeline infrastructure on **Red Hat OpenShift**, leveraging cloud-native automation to handle continuous code syncs and vector database updates.

Software Content Engineer (Werkstudent)

Jan 2026 – Feb 2026

Sliplane

Karlsruhe, Germany

- Architected and executed a technical content strategy centered on **Docker deployment** and **PaaS solutions**.
- Authored high-performing technical tutorials and "How-to" guides.

Software Engineer II

Sep 2024 – Oct 2025 | Ankara, Turkey

Lidya Genomics

- Led the development of a multi-agent pipeline for building a genetic traits knowledge base, reducing manual tasks by approximately 95%. Used LangGraph, LangChain and OpenAI APIs.

- Developed backend, frontend, and deployment features utilizing Python, Django, Django Rest Framework, React, PostgreSQL, Docker, and CI/CD.
- Worked with a serverless worker structure on Kubernetes and maintained it.
- Developed bioinformatics pipelines using Ensembl VEP, ClinVar, GATK

Software Engineer I

Jun 2023 – Sep 2024 | Ankara, Turkey

Token Financial Technologies

- Managed three critical projects focused on the administration of 1 million POS devices across Turkey, ensuring the stability and reliability of those terminals. Developed solutions by using the tech stack that includes Django, DRF, React, Oracle, PostgreSQL, RabbitMQ, Docker, and CI/CD tools.
- Built a Django and React based statistics app that improved company-wide insights and earned **3rd** place among internship projects in the intern competition.

Undergraduate Researcher, WINSLab

Feb 2024 – Jul 2024 | Ankara, Turkey

Middle East Technical University (METU)

- Developed a new dynamic transport layer protocol in the Linux kernel, a mesh networking protocol using UDP for group topology session management.

Student Assistant

Jan 2023 – Jun 2023 | Ankara, Turkey

Middle East Technical University (METU)

- Assisted in teaching Python programming to engineering students from various faculties, and prepared instructional materials.

Software Engineer - Internship

Aug 2022 – Nov 2022 | Ankara, Turkey

SNG Studios

- Developed a 2D card game from scratch using Unity and C# in collaboration with a senior developer.
- Enhanced gameplay mechanics and user experience through iterative testing and feedback.

Projects

OCR-SEC

Sep 2024 – Jul 2025

Senior Project - METU and Duke University

- Built an end-to-end OCR to LLM data pipeline to extract structured financial data from large, multi-page scanned SEC reports, combining document processing and AI orchestration into a high-accuracy, production-ready multi-agent system.

Mesh Transport Protocol (MTP)

Jan 2024 – Jul 2024

Research Project - Middle East Technical University - WiNSLab

- Researched and developed a new dynamic transport layer protocol in Linux, a mesh networking protocol using UDP for enhanced network flexibility.

Languages

English – C1, IELTS 7.5/9.0 | Turkish – Native

Skills

LangGraph | LangChain | OpenAI APIs | React | PostgreSQL | Docker | CI/CD |
 Kubernetes | Serverless Architecture | RabbitMQ | Oracle | Python | Django |
 Django Rest Framework | Distributed Computing | C++ | JavaScript | Podman | OpenShift